

[DC] Understanding the impact of the Fidelity of Multimodal Interactions in XR based Training Simulators on Cognitive Load

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ABSTRACT

As eXtended Reality (XR) technologies evolve, the integration of multimodal interfaces becomes a defining factor in immersive experiences. Existing literature highlights a lack of consensus regarding the impact of these interfaces in XR environments. Due to the Uncanny Valley Effect and its amplification in multimodal stimuli, the uncanny valley effect poses a potential challenge of increased cognitive load due to dissonance between user expectations and reality. My research pivots on the observation that current studies often overlook a crucial factor—the fidelity of stimuli presented to users. The main goal of my research is to answer the question about how multimodal interactions in XR based applications impact the cognitive load experienced by the user. To address this gap, I employ a comprehensive human-computer interaction (HCI) research approach involving frameworks, theories, user studies and guidelines. The goal is to systematically investigate the interplay of stimulus fidelity and cognitive load in XR, aiming to offer insights for the design of Audio-Visual-Haptic (AVH) interfaces.

Index Terms: Extended Reality—Multimodal Interactions—Cognitive Load—Feedback Fidelity

1 INTRODUCTION

Cognitive neuroscience and experimental psychology findings show that the brain's functioning involves multisensory perception and integration. However, studying the senses in isolation does not allow us to fully understand multisensory perception. Research from the late 1900s indicate that students learn better when using multimodal interactions [7]. This advantage spans educational activities and content domains, promoting long-term sensory-motor memory. In XR, multimodal interfaces enhance user interaction, offering a more immersive experience across diverse applications such as gaming, education, healthcare, entertainment, and collaboration [8]. In my work, I focus on AVH interfaces as numerous studies (general-purpose and task-specific studies) on AVH feedback highlight its advantages over bimodal and unimodal options [5].

Nevertheless, the literature reveals a lack of consensus, as some evidence suggests that augmenting modalities in a simulation may alter participants' responses to stimuli without significantly impacting their overall experience. Biocca et al. demonstrated that the perception of haptic feedback facilitated task execution but did not influence accuracy [2]. Contrarily, there is an indication that the addition of stimuli can negatively affect users, imposing an extra cognitive load and potentially causing overwhelm or distraction, especially when multisensory information is incongruent [1].

The literature is not in agreement about the impact of multimodal interfaces in XR environments, even though it is evident that they increase realism. This is because none of the existing studies consider the fidelity of the stimulus provided to the user. Mori stated that there is a non-linear correlation between the fidelity of the stim-

ulus and its associated human likeness called the Uncanny Valley Effect [6]. The literature also tells that the uncanny valley effect is amplified with multimodal stimuli and causes extraneous loads to working memory due to the dissonance between expectation and reality [4, 9]. Hence, it is important for XR researchers to investigate the relationship between multimodal interfaces and cognitive load through the lens of the fidelity of the input multimodal feedback to address the unique challenges posed by XR interfaces effectively.

2 RESEARCH PLAN

To address the research gap identified, I use the framework, theory, and guidelines methodology outlined in the preceding section. Initially, I describe the conceptual framework underpinning my research. Subsequently, utilizing the formulated framework, I intend to execute a user study aimed at understanding the relationship between my independent and dependent variables. Drawing upon both empirical observations and existing theoretical frameworks, I aim to construct a predictive model that provides an estimation of the subjective or perceived cognitive load encountered by the user. Finally, integrating these insights, I plan to develop design guidelines for the design and development of an AVH interface.

2.1 Part 1: Conceptual Framework for Research

In this section, I will describe the process of developing the conceptual framework for my work which helps in understanding the dynamics of my research. The independent variable for my research is the multimodal interactions experienced by the user, and the dependent variable is their reported perceived cognitive load. I moderate the relationship between the independent and dependent variables by changing the fidelity of the multimodal feedback provided to the users. One of the notable findings in the literature related to Working Memory Theory (WMT) is that expertise in a specific domain can significantly reduce cognitive load when performing domain-specific tasks. Hence, we control the participant's expertise in the task context during the recruitment of the study. Fig. 1 depicts the conceptual framework for my research. This framework serves as the anchor for our study, providing a structured framework within which I explore the dynamics of my research.

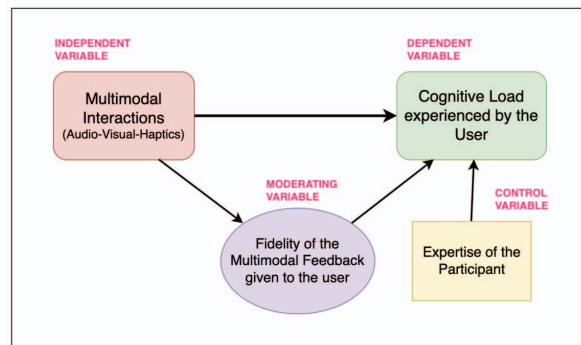


Figure 1: Conceptual framework.

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2.2 Part 2: User Study

To implement the conceptual framework, a within-participants study is planned, incorporating incremental increases in the fidelities of multimodal input for users. The study categorizes each modality into three fidelity levels: minimal/low fidelity, lacking immersion with limited real-world sensations; moderate/medium fidelity, offering a reasonable representation but lacking details; and high fidelity, closely mimicking real-world sensations for an immersive experience. These classifications draw from established terminology in literature assessing simulation fidelity and immersion levels in XR simulations. Participants will engage in the same task across all scenarios presented in random order. Data collection involves gathering behavioral data during task performance and subjective data through NASA TLX. This information will be instrumental in comprehending the cognitive load experienced by users.

Research Use Case - Human Pulse XR Simulator To conduct the study, I've developed the Human Pulse XR simulator, specifically designed for pulse palpation. It incorporates visual and auditory feedback from multi-parameter displays, along with haptic feedback from palpation. Each feedback mode conveys unique information about specific pulse parameters; for example, auditory cues relay information about blood oxygen saturation, while haptic feedback conveys data on pulse pressure and rate. Participants use these modalities to effectively categorize pulses. The simulator, presented in a publication at an IEEE VR workshop in 2023 [3], proves ideal for our study, given that the task of palpating pulses demands the integration of information from all three feedback modalities.

2.3 Part 3: Design of Predictive Model

Building upon the insights derived from our study and leveraging existing predictive models, my goal is to construct a predictive model. This model will utilize input parameters encompassing the categorized fidelity levels of visual, auditory, and haptic feedback to forecast the anticipated subjective cognitive load.

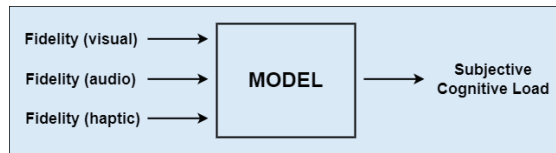


Figure 2: Schematic diagram of the predictive model to be developed.

Fig. 2 provides a visual representation of the proposed model. To substantiate the efficacy and fine-tune the formulated model, I plan to conduct user studies on two distinct XR-based training simulators focused in the medical field where the cognitive load experienced is high. Both the training simulators will be designed to include AVH interfaces to help perform the tasks. The study design will be guided by the developed model where we will monitor the change in subjective cognitive load due to change in the input fidelity and compare it with the results predicted by the model.

2.4 Part 4: Design Guidelines

Numerous theoretical frameworks, including Multiple Resource theory and Cognitive Load theory, have significantly influenced the design of multimodal interfaces, serving as conceptual foundations for the development of more robust systems. The amalgamation of different modalities, when appropriately integrated, proves instrumental in conveying information to users more effectively. This approach leverages various components of working memory, thereby alleviating the cognitive load on any individual element.

Interestingly, while existing design guidelines draw upon these theories, none specifically guide XR designers in aligning their designs with the requisite cognitive load for a given application. In

response to this gap, I intend to propose design guidelines that empower XR application developers to judiciously modulate modality feedback, achieving the desired cognitive load. These guidelines will be laid by the observations made in the study and the predictive model developed earlier.

3 CONCLUSION

In conclusion, the research delves into the interplay of multimodal interactions, stimulus fidelity, and cognitive load in XR environments. Yet, defining fidelity and determining validation methodologies for predictive models present two pivotal challenges. The first challenge revolves around striking a balance between objective parameters and subjective user perceptions when defining fidelity. The second challenge involves exploring alternative validation methods beyond user studies to enhance the robustness and generalizability of our predictive model. Meeting these challenges is vital for the success of my research in the evolving landscape of extended reality. I am currently in my first year of PhD and my participation in the Doctoral Consortium will empower me to navigate the complexities of my Ph.D. journey with greater confidence.

4 IMPORTANCE

The significance of this research lies in its pivotal role in navigating the complexities of multimodal interfaces within XR environments. With technological advancements in immersive experiences, understanding the interplay of auditory, visual, and haptic stimuli becomes paramount. The outcomes of this research hold promise for the academic discourse on human-computer interaction and for practical applications in XR design. The proposed framework, user study, and predictive model aim to explore the intricacies of how different levels of fidelity in multimodal feedback impact cognitive load. This, in turn, can inform the development of nuanced design guidelines, empowering XR designers to craft interfaces that strike an optimal balance between realism and cognitive load. As XR technologies continue to proliferate across diverse domains such as education and healthcare, the findings of this research bear implications for enhancing user experience, task performance, and overall usability.

REFERENCES

- [1] M. Bessa, M. Melo, A. A. De Sousa, and J. Vasconcelos-Raposo. The effects of body position on reflexive motor acts and the sense of presence in virtual environments. *Computers & Graphics*, 71:35–41, 2018.
- [2] F. Biocca, J. Kim, and Y. Choi. Visual touch in virtual environments: An exploratory study of presence, multimodal interfaces, and cross-modal sensory illusions. *Presence: Teleoperators & Virtual Environments*, 10(3):247–265, 2001.
- [3] N. D. Chandrashekar, S. Safford, M. Muniyandi, and D. Gračanin. An extended reality simulator for pulse palpation training. In *2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 178–182. IEEE, 2023.
- [4] M. C. Howard. Investigating the simulation elements of environment and control: Extending the uncanny valley theory to simulations. *Computers & Education*, 109:216–232, 2017.
- [5] M. Melo, G. Gonçalves, P. Monteiro, H. Coelho, J. Vasconcelos-Raposo, and M. Bessa. Do multisensory stimuli benefit the virtual reality experience? a systematic review. *IEEE Transactions on Visualization and Computer Graphics*, 28(2):1428–1442, 2020.
- [6] M. Mori, K. F. MacDorman, and N. Kageki. The uncanny valley [from the field]. *IEEE Robotics & automation magazine*, 19(2):98–100, 2012.
- [7] S. Y. Mousavi, R. Low, and J. Sweller. Reducing cognitive load by mixing auditory and visual presentation modes. *Journal of educational psychology*, 87(2):319, 1995.
- [8] S. Philippe, A. D. Souchet, P. Lameris, P. Petridis, J. Caporal, G. Coldebœuf, and H. Duzan. Multimodal teaching, learning and training in virtual reality: a review and case study. *Virtual Reality & Intelligent Hardware*, 2(5):421–442, 2020.
- [9] E. Yorgancıgil, B. A. Urgan, and F. Yildirim. Uncanny valley effect is amplified with multimodal stimuli and varies across ages. 2021.